Phys. Fluids **35** 012114

44. Denner F 2024 The Kirkwood–Bethe hypothesis for bubble dynamics, cavitation, and underwater explosions *Phys. Fluids* **36** 051302
45. Lauterborn W and Bolle H 1975 Experimental Investigations of Cavitation-Bubble Collapse in Neighborhood of a Solid Boundary *J. Fluid. Mech.* **72** 391-399
46. Vogel A, Lauterborn W, and Timm R 1989 Optical and acoustic investigations of the dynamics of laser-produced cavitation bubbles near a solid boundary *J. Fluid. Mech.* **206** 299-338
47. Juhasz T, Kastis G A, Suarez C, Bor Z, and Bron W E 1996 Time-resolved observations of shock waves and cavitation bubbles generated by femtosecond laser pulses in corneal tissue and water *Laser Surg. Med.* **19** 23-31
48. Philipp A and Lauterborn W 1998 Cavitation erosion by single laser-produced bubbles *J. Fluid. Mech.* **361** 75-116
49. Ohl C D, Kurz T, Geisler R, Lindau O, and Lauterborn W 1999 Bubble dynamics, shock waves and sonoluminescence *Phil. Trans. R. Soc. A* **357** 269-294
50. Brujan E A, Nahen K, Schmidt P, and Vogel A 2001 Dynamics of laser-induced cavitation bubbles near an elastic boundary *J. Fluid. Mech.* **433** 251-281
51. Lindau O and Lauterborn W 2003 Cinematographic observation of the collapse and rebound of a laser-produced cavitation bubble near a wall *J. Fluid. Mech.* **479** 327-348
52. Obreschkow D, Tinguely M, Dorsaz N, Kobel P, de Bosset A, and Farhat M 2013 The quest for the most spherical bubble: experimental setup and data overview *Exp. Fluids.* **54** 1503
53. Supponen O, Obreschkow D, Tinguely M, Kobel P, Dorsaz N, and Farhat M 2016 Scaling laws for jets of single cavitation bubbles *J. Fluid. Mech.* **802** 263-293
54. Sinibaldi G, Occhicone A, Pereira F A, Caprini D, Marino L, Michelotti F, and Casciola C M 2019 Laser induced cavitation: Plasma generation and breakdown shockwave *Phys. Fluids* **31** 103302
55. Gonzalez-Avila S R, Denner F, and Ohl C-D 2021 The acoustic pressure generated by the cavitation bubble expansion and collapse near a rigid wall *Phys. Fluids* **33** 032118
56. Reuter F, Deiter C, and Ohl C-D 2022 Cavitation erosion by shockwave self-focusing of a single bubble *Ultrason. Sonochem.* **90** 106131
57. Rosselló J M, Reese H, and Ohl C-D 2022 Dynamics of pulsed laser-induced cavities on a liquid–gas interface: from a conical splash to a 'bullet' jet *J. Fluid. Mech.* **939** A35
58. Cole R H 1948 *Underwater Explosions* (Pinceton University Press)
59. Sternberg H M and Walker W A 1971 Calculated Flow and Energy Distribution Following Underwater Detonation of a Pentolite Sphere *The Physics of Fluids* **14** 1869-1878
60. Tomita Y and Shima A 1986 Mechanisms of Impulsive Pressure Generation and Damage Pit Formation by Bubble Collapse *J. Fluid. Mech.* **169** 535-564
61. Podbevšek D, Lokar Ž, Podobnikar J, Petkovšek R, and Dular M 2021 Experimental evaluation of methodologies for single transient cavitation bubble generation in liquids *Exp. Fluids.* **62** 167
62. Akhatov I, Lindau O, Topolnikov A, Mettin R, Vakhitova N, and Lauterborn W 2001 Collapse and rebound of a laser-induced cavitation bubble *Phys. Fluids* **13** 2805-2819
63. Lauterborn W and Kurz T 2010 Physics of bubble oscillations *Rep. Prog. Phys.* **73** 106501
64. Koch M, Lechner C, Reuter F, Kohler K, Mettin R, and Lauterborn W 2016 Numerical modeling of laser

Accepted Manuscript

generated cavitation bubbles with the finite volume and volume of fluid method, using OpenFOAM *Comput. Fluids* **126** 71-90

65. Lechner C, Lauterborn W, Koch M, and Mettin R 2020 Jet formation from bubbles near a solid boundary in a compressible liquid: Numerical study of distance dependence *Phys Rev Fluids* **5** 093604
66. Wen H, Yao Z, Zhong Q, Tian Y, Sun Y, and Wang F 2023 Energy partitioning in laser-induced millimeter-sized spherical cavitation up to the fourth oscillation *Ultrason. Sonochem.* **95** 106391
67. Bußmann A, Riahi F, Gökce B, Adami S, Barcikowski S, and Adams N A 2023 Investigation of cavitation bubble dynamics near a solid wall by high-resolution numerical simulation *Phys. Fluids* **35** 016115
68. Zhang A M, Li S-M, Xu R-Z, Pei S-C, Li S, and Liu Y-L 2024 A theoretical model for compressible bubble dynamics considering phase transition and migration *J. Fluid. Mech.* **999** A58
69. Kabashin A V and Meunier M 2003 Synthesis of colloidal nanoparticles during femtosecond laser ablation of gold in water *J. Appl. Phys.* **94** 7941-7943
70. Sylvestre J P, Kabashin A V, Sacher E, and Meunier M 2005 Femtosecond laser ablation of gold in water: influence of the laser-produced plasma on the nanoparticle size distribution *Appl. Phys. A* **80** 753-758
71. Dell'Aglio M, Gaudiuso R, De Pascale O, and De Giacomo A 2015 Mechanisms and processes of pulsed laser ablation in liquids during nanoparticle production *Appl. Surf. Sci.* **348** 4-9
72. Zhang D, Gökce B, and Barcikowski S 2017 Laser Synthesis and Processing of Colloids: Fundamentals and Applications *Chem. Rev.* **117** 3990-4103
73. Streubel R, Barcikowski S, and Gökce B 2016 Continuous multigram nanoparticle synthesis by high-power, high-repetition-rate ultrafast laser ablation in liquids *Opt. Lett.* **41** 1486-1489
74. Barcikowski S, Plech A, Suslick K S, and Vogel A 2019 Materials synthesis in a bubble *Mrs Bull* **44** 382-391
75. Kanitz A, Kalus M R, Gurevich E L, Ostendorf A, Barcikowski S, and Amans D 2019 Review on experimental and theoretical investigations of the early stage, femtoseconds to microseconds processes during laser ablation in liquid-phase for the synthesis of colloidal nanoparticles *Plasma Sources Sci T* **28** 103001
76. Dell'Aglio M, Motto-Ros V, Pelascini F, Gornushkin I B, and De Giacomo A 2019 Investigation on the material in the plasma phase by high temporally and spectrally resolved emission imaging during pulsed laser ablation in liquid (PLAL) for NPs production and consequent considerations on NPs formation *Plasma Sources Science and Technology* **28** 085017
77. Fazio E, et al. 2020 Nanoparticles Engineering by Pulsed Laser Ablation in Liquids: Concepts and Applications *Nanomaterials* **10** 2317
78. Dell'Aglio M, Gardette V, Jantzi S C, and De Giacomo A 2021 Comparison between laser induced plasmas in gas and in liquid *J. Appl. Phys.* **129** 233303
79. Shaheen M E and Abdelwahab A Y E 2025 Laser ablation in liquids: A versatile technique for nanoparticle generation *Optics & Laser Technology* **186** 112705
80. Wu J, Zhou Y, Chen M, and Li X 2025 Numerical modeling and simulation on nanosecond laser-target interactions *Journal of Physics D: Applied Physics* **58** 143004
81. Shih C-Y, Shugaev M V, Wu C, and Zhigilei L V 2020 The effect of pulse duration on nanoparticle generation in pulsed laser ablation in liquids: insights from large-scale atomistic simulations *Phys. Chem. Chem. Phys.* **22** 7077-7099
82. Cremers D A, Radziemski L J, and Loree T R 1984 Spectrochemical Analysis of Liquids Using the Laser Spark *Appl Spectrosc* **38** 721-729

Accepted Manuscript

83. Charfi B and Harith M A 2002 Panoramic laser-induced breakdown spectrometry of water *Spectrochimica Acta Part B: Atomic Spectroscopy* **57** 1141-1153

84. Hahn D W and Omenetto N 2010 Laser-Induced Breakdown Spectroscopy (LIBS), Part I: Review of Basic Diagnostics and Plasma-Particle Interactions: Still-Challenging Issues Within the Analytical Plasma Community *Appl Spectrosc* **64** 335a-366a

85. Thornton B, Takahashi T, Sato T, Sakka T, Tamura A, Matsumoto A, Nozaki T, Ohki T, and Ohki K 2015 Development of a deep-sea laser-induced breakdown spectrometer for in situ multi-element chemical analysis *Deep Sea Research Part I: Oceanographic Research Papers* **95** 20-36

86. Angel S M, Bonvallet J, Lawrence-Snyder M, Pearman W F, and Register J 2016 Underwater measurements using laser induced breakdown spectroscopy *Journal of Analytical Atomic Spectrometry* **31** 328-336

87. López-Claros M, Dell'Aglio M, Gaudiuso R, Santagata A, De Giacomo A, Fortes F J, and Laserna J J 2017 Double pulse laser induced breakdown spectroscopy of a solid in water: Effect of hydrostatic pressure on laser induced plasma, cavitation bubble and emission spectra *Spectrochimica Acta Part B: Atomic Spectroscopy* **133** 63-71

88. Jia Z, et al. 2023 Dynamics of laser-induced plasma and cavitation bubble at high pressures and the impacts on underwater LIBS signals *Spectrochimica Acta Part B: Atomic Spectroscopy* **209** 106793

89. Takahashi T, Yoshino S, Takaya Y, Nozaki T, Ohki K, Ohki T, Sakka T, and Thornton B 2020 Quantitative in situ mapping of elements in deep-sea hydrothermal vents using laser-induced breakdown spectroscopy and multivariate analysis *Deep Sea Research Part I: Oceanographic Research Papers* **158** 103232

90. Ibrahimkutty S, Wagener P, Rolo T D, Karpov D, Menzel A, Baumbach T, Barcikowski S, and Plech A 2015 A hierarchical view on material formation during pulsed-laser synthesis of nanoparticles in liquid *Sci. Rep.* **5** 163131-163111

91. Reich S, et al. 2017 Pulsed laser ablation in liquids: Impact of the bubble dynamics on particle formation *Journal of Colloid and Interface Science* **489** 106-113

92. Reich S, Schönfeld P, Letzel A, Kohsakowski S, Olbinado M, Gökce B, Barcikowski S, and Plech A 2017 Fluence Threshold Behaviour on Ablation and Bubble Formation in Pulsed Laser Ablation in Liquids *ChemPhysChem* **18** 1084-1090

93. Kalus M-R, Bärsch N, Streubel R, Gökce E, Barcikowski S, and Gökce B 2017 How persistent microbubbles shield nanoparticle productivity in laser synthesis of colloids – quantification of their volume, dwell dynamics, and gas composition *Phys. Chem. Chem. Phys.* **19** 7112-7123

94. Shih C-Y, et al. 2018 Two mechanisms of nanoparticle generation in picosecond laser ablation in liquids: the origin of the bimodal size distribution *Nanoscale* **10** 6900-6910

95. Takami A, Kurita H, and Koda S 1999 Laser-Induced Size Reduction of Noble Metal Particles *J. Phys. Chem. B* **103** 1226-1232

96. Amendola V and Meneghetti M 2007 Controlled size manipulation of free gold nanoparticles by laser irradiation and their facile bioconjugation *Journal of Materials Chemistry* **17** 4705

97. Huang H and Zhigilei L V 2021 Atomistic View of Laser Fragmentation of Gold Nanoparticles in a Liquid Environment *The Journal of Physical Chemistry C* **125** 13413-13432

98. Frias Batista L M, Nag A, Meader V K, and Tibbetts K M 2022 Generation of nanomaterials by reactive laser-synthesis in liquid *Science China Physics, Mechanics & Astronomy* **65** 274202

99. Huang H and Zhigilei L V 2022 Computational study of laser fragmentation in liquid: Phase explosion, inverse Leidenfrost effect at the nanoscale, and evaporation in a nanobubble *Science China Physics,*

Accepted Manuscript